



DEPARTAMENTO
DE INFORMÁTICA
PUC-RIO

Introduction

Perspectives on agent systems



Outline

- **Motivation**
- Agent and Multiagent Systems, a definition
- Examples of agent-based systems



Motivation

- Four **ongoing trends** have marked the history of computing:
 - *ubiquity;*
 - *interconnection;*
 - *intelligence;*
 - *delegation.*

Motivation



Ubiquity



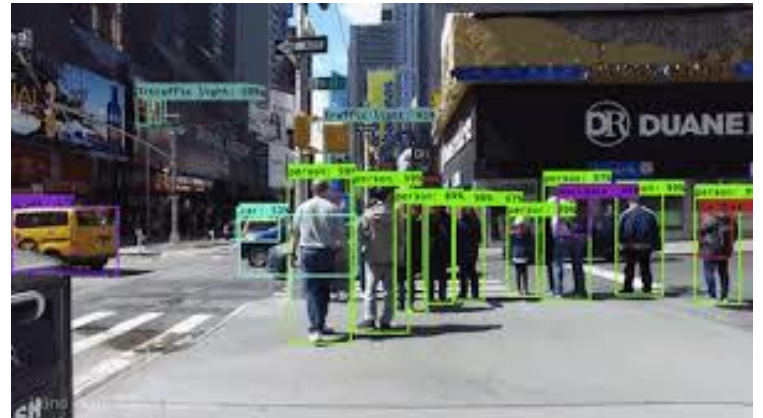
Motivation



Interconnection



Motivation



Artificial Intelligence



Motivation



Delegation



Where does it bring us?

- All of these trends have led to the emergence of a new field in Computer Science: *multiagent systems*

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- **Agent and Multiagent Systems, a definition**
- Examples of agent-based systems



Agents, a Definition

- An **agent** is a **computer system** that is capable of *independent* (*autonomous*) action on behalf of its user or owner

Agents, a Definition

- *Autonomy* is a key aspect of an agent!
 - An agent should figure out what to do
 - Rather than constantly being told what to do

Multiagent Systems, a Definition

- A multiagent system is one that consists of a *number of agents*, which *interact* with one another
- In the most general case, agents will be *acting on behalf of users with different goals and motivations*
- To successfully interact, they will require the ability to *cooperate*, *coordinate*, and *negotiate* with each other, much as people do

The two key problems

- *Agent design: How do we build agents?*
 - Agents that are independent, autonomous, and able to carry out tasks we delegate to them
 - Decision making is a key aspect!

The two key problems

- *Society design: How do we build agents that are capable of interacting?*
 - cooperating, coordinating, negotiating with other agents
- These are the *micro* and *macro* perspectives

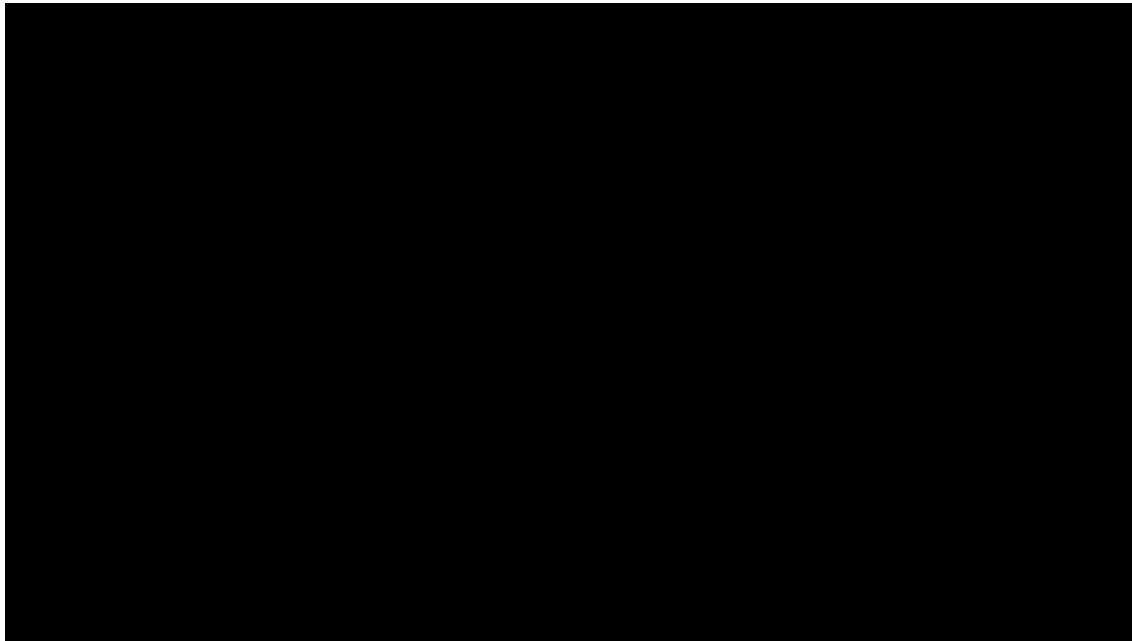
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- **Examples of agent-based systems**



Examples of Agents

- *Autonomy - NASA's Mars 2021 Perseverance rover*



<https://www.youtube.com/watch?v=M4tdMR5HLtg>

Examples of Agents

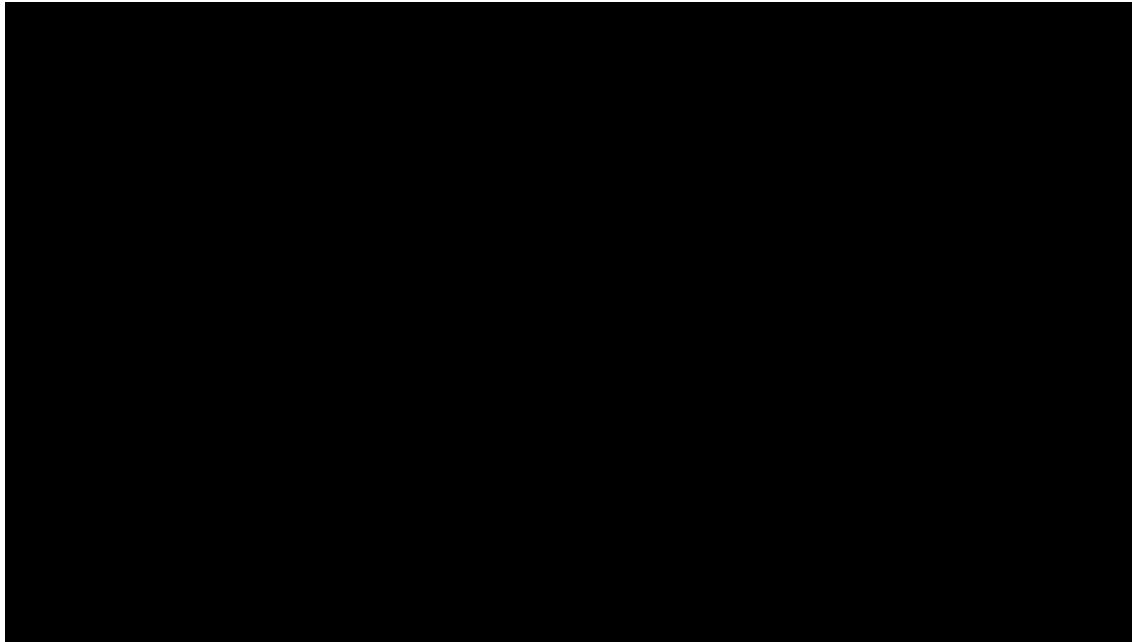
- *Cooperation and Coordination - RoboCup*



https://www.youtube.com/watch?v=_Y5_iGxWFrQ

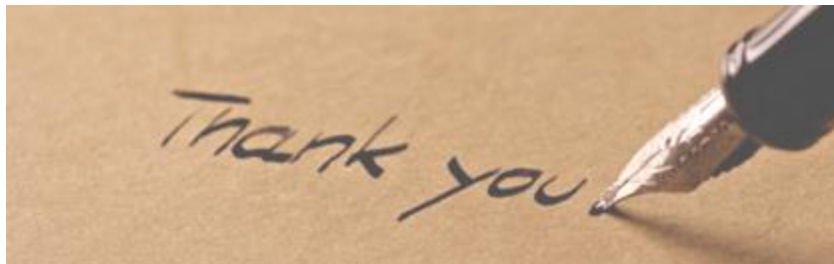
Examples of Agents

- *Trading Agent – Automated/Algorithmic Trading*



https://www.youtube.com/watch?v=OPm_EDTrz7Y

Thank You



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